

Question 1: a description and reflection of the problem your team is interested in addressing

Currently, there's a whole market for the best ways to optimize credit card points. There are many people who are *paid* to look through thousands of websites and to find which conversion has a current discount, which flight is offered that can boost flight points, which hotel is undervalued and free through which credit card union that has a current free trial going. Not only does this mean the already rich can pay others to find the best way to save (or even make more) money, everyone can agree the convoluted part-time/full-time offers of credit cards, flight companies, hotels, and others are extremely annoying. For most people, expending the effort to find the *best* offer is just too much effort. For some, a good enough offer is fine, for others, there's not even enough time to make looking for any offers worth it because of the psychological toll decision paralysis could play on them.

Outside of just time and money, expertise is required. Just like how many online scams can be easily run with existing money and the knowledge of how they work, point conversion has become an expertise. This is done on purpose by the many connected companies because they make money when people don't find the most optimal way to use their points. With all the existing conditions around this problem, the people who lose money are the ones who need it the most.

It's also rather ridiculous that one of the most basic financial decisions in the US requires so much thought. With how normalized credit cards are in the US, deciding on a credit card means first deciding on a debit card, on a bank, on a level. It means balancing cashback, flight points, restaurant offers, hotels, annual fees and more to decide on a card that is best for each specific person. All of this information is public, and with the advent of AI searching through all this information has become much simpler, but it could be improved even more.

Question 2: a description and reflection on the audience the team is interested in supporting

For the people who want to save points and money (everyone), for the people who want to fly and vacation cheaper (most people), or for the people who use credit so much, simplifying the process is already easing a pain point. The exact audience would depend on how specified the scope of the project is, and in which direction it is specified.

If, for example, I focused on starter credit cards, the audience would be international students and young adults. Some people's parents got their credit started early by adding them on companion cards, so that they would have more options and qualify for more when they come of age, but others (like mine) didn't do that. I, personally, applied to at least three different credit cards when I turned eighteen and was flatly rejected. Having a tool that could help you pick out cards and not waste your time, as well as help guide you through the process would be massively beneficial. Questions like income, allowance, or annual payments usually don't have a clear answer for someone who just turned eighteen, so having something to help you with it, especially as an international student, could save people a lot of time and effort.

As another example, focusing the scope onto credit point transfers would mean the audience would be most of the middle class. People who have jobs, who have multiple credit cards, and who aren't utilizing them to their fullest. Having a tool that can alert you when there's a special deal, or quickly find which card already has an offer, would be like a discount version of a personal advisor.

A third possibility is focusing specifically on credit point to flight point transfers. This would work best for people who fly often. Which flight path would be cheapest? Would utilize points most effectively? Which path might be slightly more expensive but allows rest time and

free lounge use? Hotel memberships? Again, all this information is public, it is simply that most people don't have the time and energy to find all of it.

Question 3: approaches/ideas/paths the team is interested in exploring

My current idea is an LLM assisted online scraper. Since all the information is public, getting and compiling all of it is the first step. Because of the many antiscraper bots, the LLM would be assisting in this step to read off of the webpage.

The next step would be to either show that information on a webpage or give it to some tool directly, so that all of that information can be found in one place. The TrumpTracker comes to mind, because it accurately displays a lot of information but is quite boring to go through and looks like a statistical tool.

The next idea is to have a language based LLM trained on all of that information (frequently renewed of course) so that people could prompt and receive information instead of going hunting for it (even if it's all in one place).

Going in a completely different direction, instead of creating a tool that collects all of that information, maybe a collection of educational sites could also help. Instead of providing answers directly, provide a list where the answers would be. I'm not fully happy with this sort of solution, but it is the easiest to make.

The last idea I currently have is instead some sort of collaboration internationally where international credit history can be transferred. This is a long shot but sometimes I like to imagine all that data can be responsibly handled in a way that benefits everyone.

Question 4: an overview of the team members' personal interest and background

I (Ava Cheng) have a varied and foundational history of work experience, and I believe these diverse experiences combined with my strong analytical foundation and programming experiences make me a strong candidate. In the past, I applied and was offered an internship as a marketing analyst at a social media entertainment startup called Mideo. The company was interested in my generation's internet and media habits, so I did independent research on past marketing strategies and compiled a presentation with my findings. I also assisted with product fit and design, both with my own opinions and through creation and promotion of surveys. My second internship was with Lentera Anak Pelangi, UNAIDS partnered non-profit based in Indonesia, as a Data Analyst Intern. My manager was generally inaccessible because of travel and I had to teach myself how to program equations that could calculate HIV risk levels for vulnerable children. I conducted longitudinal data analysis, gaining hands-on experience managing and analyzing real-world datasets.

Since starting my first year of college, I've worked on a couple of different projects. With a group of friends, we competed in an MIT Green Energy Hackathon. Our project centered around matching waste producers to waste consumers in an attempt to algorithmically find the most optimal strategy to reduce waste without overly demanding change from companies and factories themselves. I have also worked on a couple personal projects. In one, I played with machine learning and attempted to create an AI that could accurately convert numbers written in text (ie two) to numerical numbers (ie 2). In another, I used python to convert complex 3D mathematical equations into an STL file.

I have also joined several clubs, and have been elected by the freshman class to work on the Student Honor Board as a freshman representative. My two favorite clubs currently are the

Mudd Amateur Rocketry Club (MARC) and the MuddEscapes club. In MARC, I'm working on the Avionics team on designing our own flight black boxes for storing flight information and rocket retrieval. In MuddEscapes, I worked with a small group to design and create a roulette wheel combination lock that opened one door of the escape room.